

67. Figure 24.45 shows several point charges and several surfaces which enclose some of these charges. For each of the surfaces, what is the electric flux?

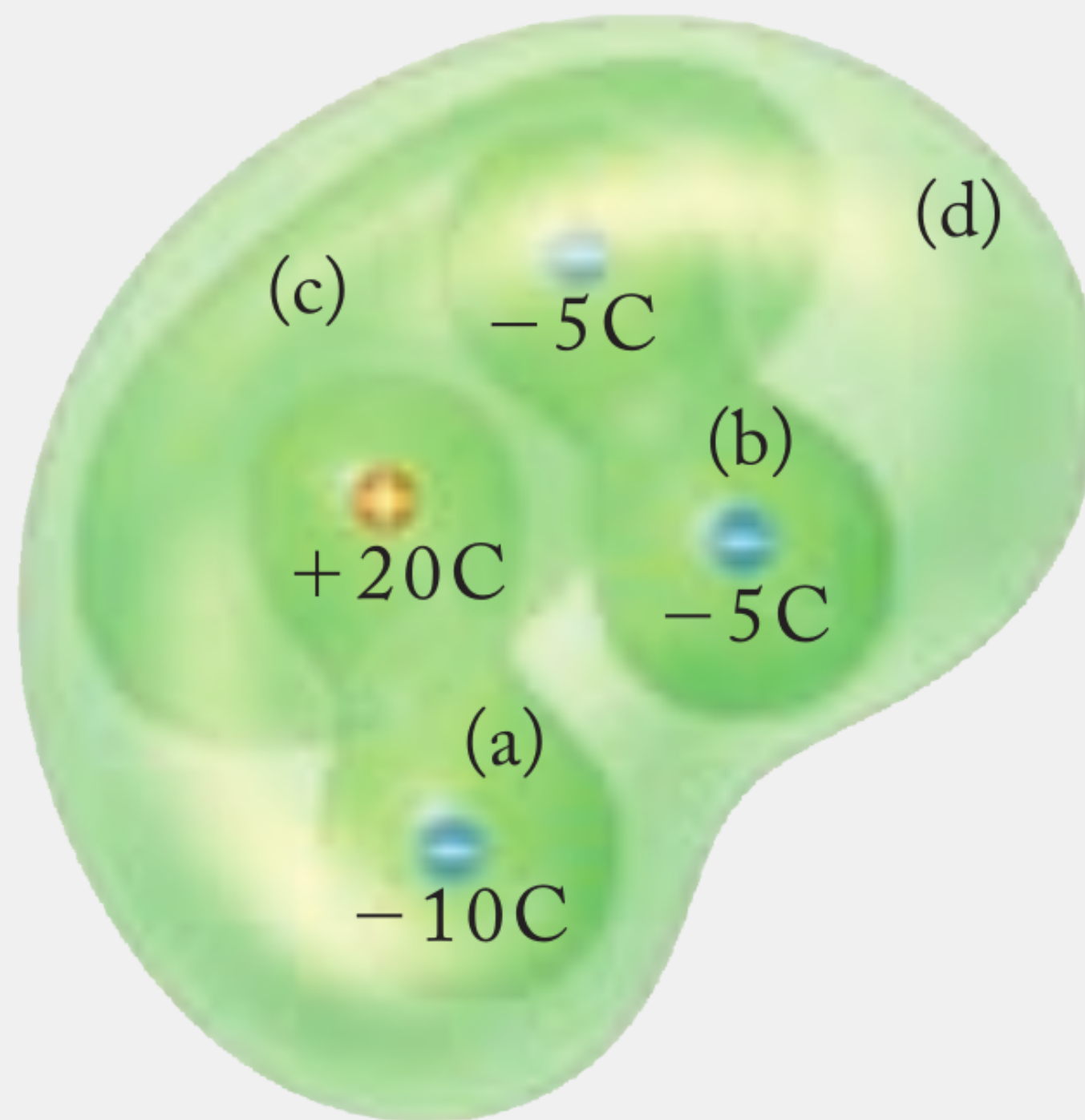
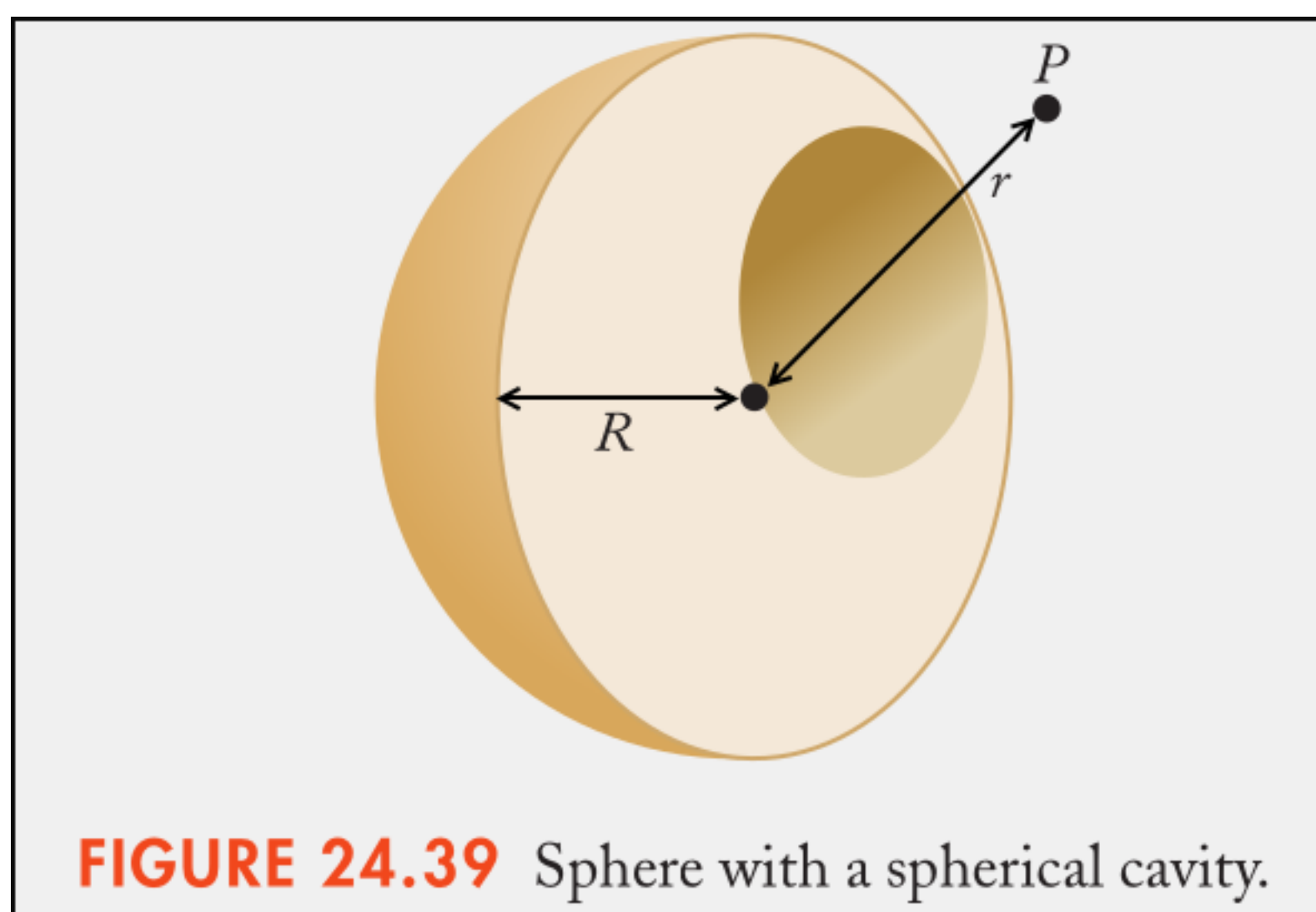


FIGURE 24.45 Several Gaussian surfaces (a–d).

Positive charge Q is uniformly distributed over the volume of a solid sphere of radius R . Suppose that a spherical cavity of radius $R/2$ is cut out of the solid sphere, the center of the cavity being at a distance of $R/2$ from the center of the original solid sphere (Fig. 24.39); the cut-out material and its charge are discarded. What new electric field does the sphere with the cavity produce at the point P at a distance r from the original center? Assume $r > R$.



*36. According to a (crude) model, the neutron consists of an inner core of positive charge surrounded by a shell of negative charge. Suppose that the positive charge has a magnitude $+e$ and is uniformly distributed over a sphere of radius 0.50×10^{-15} m; suppose that the negative charge has a magnitude $-e$ and is uniformly distributed over a concentric shell of inner radius 0.50×10^{-15} m and outer radius 1.0×10^{-15} m (Fig. 24.36). Find the magnitude and direction of the electric field at 1.0×10^{-15} , 0.75×10^{-15} , 0.50×10^{-15} , and 0.25×10^{-15} m from the center.

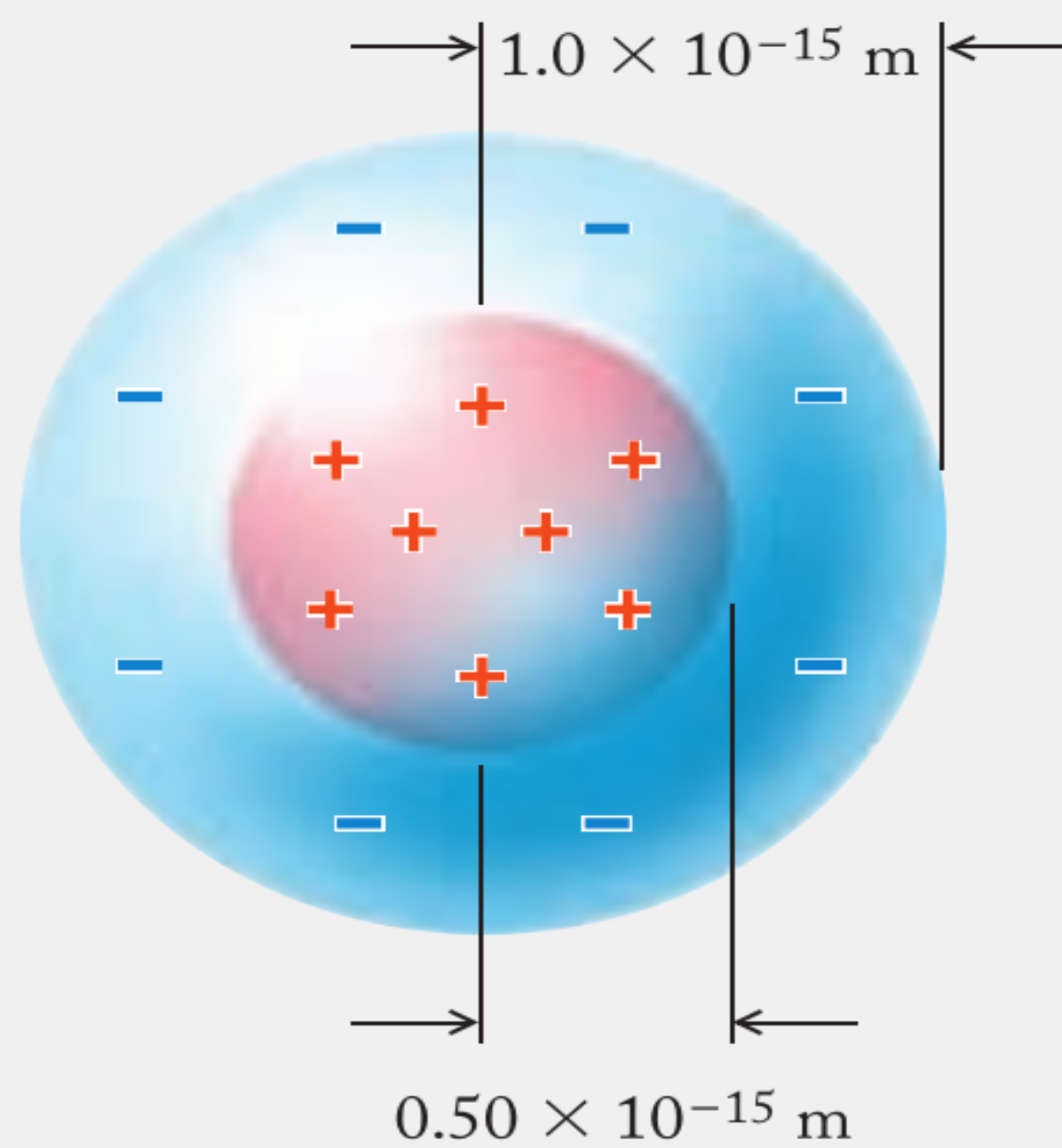


FIGURE 24.36 Hypothetical charge distribution inside a neutron.

- *33. A plastic spherical shell has inner radius a and outer radius b (Fig. 24.33). Electric charge is uniformly distributed over the region $a < r < b$. The amount of charge is ρ coulombs per cubic meter. Find the electric field in the regions $r \leq a$, $a \leq r \leq b$, and $r \geq b$.

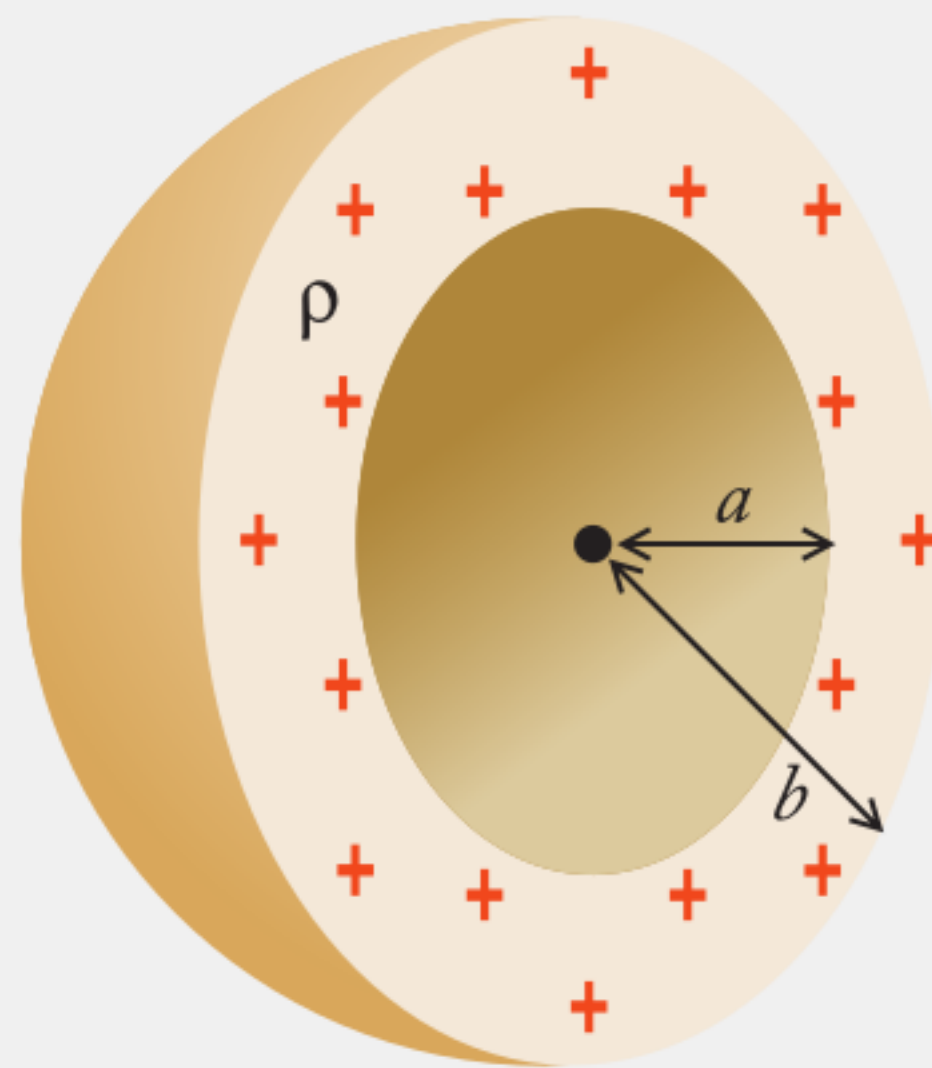


FIGURE 24.33 A spherical shell with uniform volume charge density ρ .

- 32.** Assume the magnitude of the electric field on each face of the cube of edge $L = 1.00$ m in Figure P24.32 is uniform and the directions of the fields on each face are as indicated. Find (a) the net electric flux through the cube and (b) the net charge inside the cube. (c) Could the net charge be a single point charge?

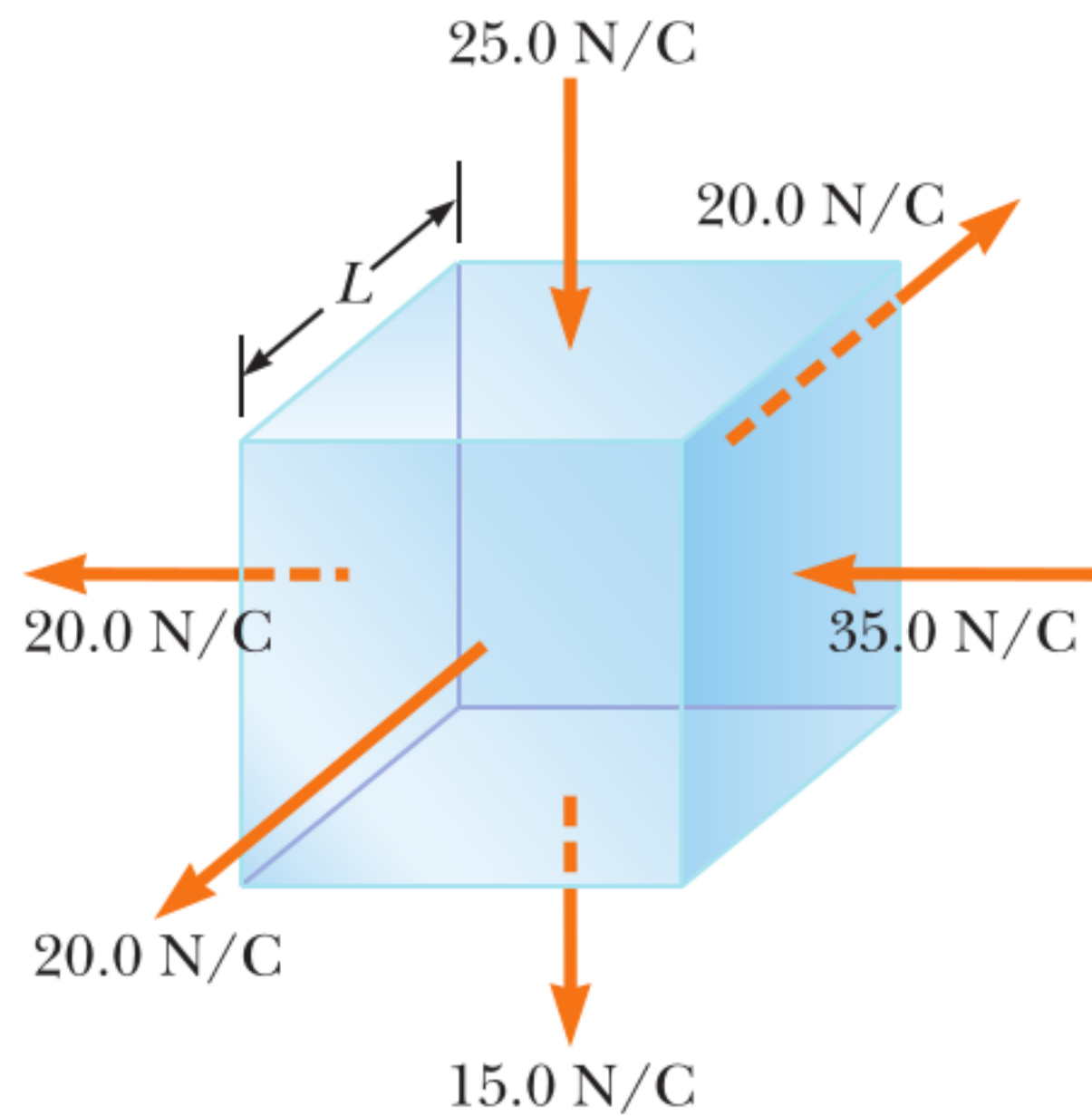


Figure P24.32

68. A particle with charge Q is located on the axis of a circle of radius R at a distance b from the plane of the circle (Fig. P24.68). Show that if one-fourth of the electric flux from the charge passes through the circle, then $R = \sqrt{3}b$.

